2026年10月16日

Excellent

Page 101, In Class Exercise 4.2.1, Page 118 A (1)(2)(3)(6)(12), Page 119 B (1)(2)(4)(8)

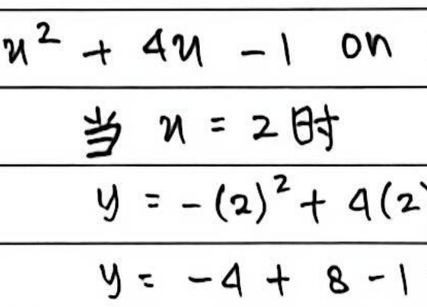
Page 101 (Quadratic Functions)

(1) Find the opening direction, axis of symmetry and vertex of $y = 3x^2 - 9x + 1$

→ Opening direction open upward (开口向上)

• Axis of symmetry	• Vertex (顶点)
$x = \frac{-b}{2a}$	$x_p = \frac{-b}{2a} = \frac{3}{2}$
$x = \frac{-(-9)}{2(3)}$	$y_p = \frac{b^2 - 4ac}{-4a} = \frac{(-9)^2 - 4(3)(1)}{-4(3)} = \frac{81-12}{-12}$
$x = \frac{9}{6}$ $x = \frac{3}{2}$ //	$= \frac{69}{-12} = -\frac{23}{4}$ //

Vertex $= (\frac{3}{2}, -\frac{23}{4})$ //

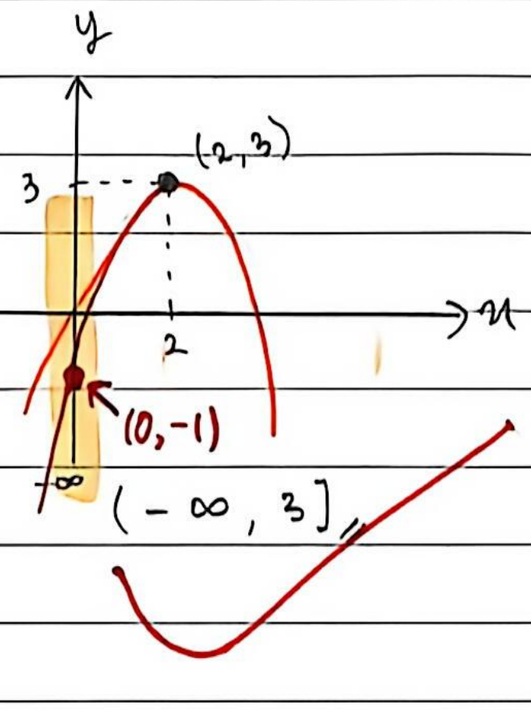


(2) Find the range of $y = -x^2 + 4x - 1$ on $\mathbb{R}$

→ $x_p = \frac{-b}{2a} = \frac{-4}{2(-1)} = 2$ //

$y_p = \frac{b^2 - 4ac}{-4a} = \frac{(4)^2 - 4(-1)(-1)}{-4(-1)}$

$= \frac{16 - 4}{4} = 3$ //

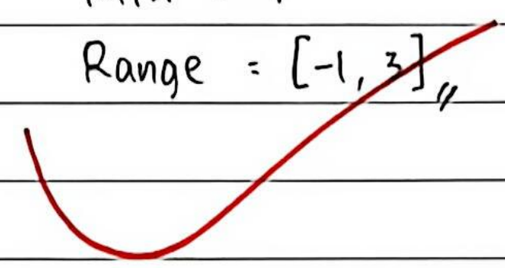


$(-\infty, 3]$ //

(3) Find the range of $y = -x^2 + 4x - 1$ on $[0, 3]$

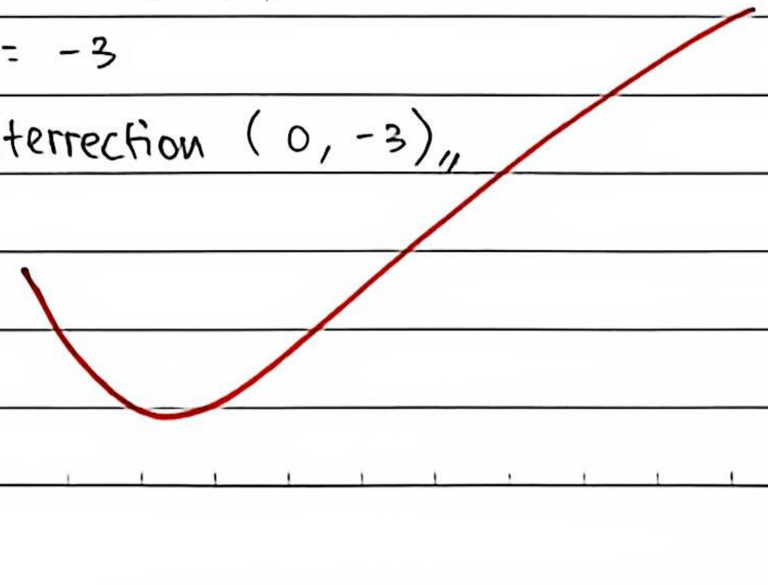
→ 当 $x = 0$ 时 当 $x = 2$ 时

$y = -0^2 + 4(0) - 1$	$y = -(2)^2 + 4(2) - 1$	Max = 3
$y = -1$ //	$y = -4 + 8 - 1$ 、$y = 3$ //	Min = -1
当 $x = 1$ 时	当 $x = 3$ 时	Range = $[-1, 3]$ //
$y = -(1)^2 + 4(1) - 1$	$y = -(3)^2 + 4(3) - 1$	
$y = 2$ //	$y = -9 + 12 - 1$ $y = 2$ //	



(4). Find the intersections of the parabola $y = x^2 + 2x - 3$ with the coordinate axes

→ Intersections with x-axes, $y = 0$	Intersection with y-axes, $x = 0$
$0 = x^2 + 2x - 3$	$y = 0^2 + 2(0) - 3$
$0 = (x - 1)(x + 3)$	$y = -3$
$x - 1 = 0$ ∨ $x + 3 = 0$	Intersection $(0, -3)$ //
$x = 1$ // $x = -3$ //	
Intersection $(1, 0)$ & $(-3, 0)$ //	



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(5). A quadratic function has zeros 1, 3 (x_1, x_2), and passes through (0, 3). Find its analytic expression.

→ $f(x) = a(x - x_1)(x - x_2)$

$= a(x - 1)(x - 3)$

Since $f(0) = 3$ substitute $x = 0$, $f(x) = 3$

↝ $3 = a(0 - 1)(0 - 3)$

$3 = a(-1)(-3)$

$3 = 3a$

$a = \frac{3}{3} = 1$ //

$a = 1$

↳ $f(x) = (x - 1)(x - 3)$

$f(x) = x^2 - 3x - x + 3$

$f(x) = x^2 - 4x + 3$ //

The analytic expression is $f(x) = x^2 - 4x + 3$ //

(6). Given that $y = ax^2 + bx - 3$ passes through $(-4, 5)$ (x, y) and has axis of symmetry $x = -1$, find a, b.

→ $x = \frac{-b}{2a}$

$-1 = \frac{-b}{2a}$

multiply both sides by $2a$

↳ $-2a = -b$

$b = 2a$ //

$y = ax^2 + bx - 3$

$5 = a(-4)^2 + b(-4) - 3$

$5 = 16a - 4b - 3$

$5 + 3 = 16a - 4b$

$8 = 16a - 4b$ ÷4

$2 = 4a - b$ //

↝ $2 = 4a - b$

$2 = 4a - 2a$

$2 = 2a$

$a = \frac{2}{2}$

$a = 1$ //

$b = 2a$

$b = 2(1)$

$b = 2$

$a = 1, b = 2$ ↝ $y = x^2 + 2x - 3$ //

Page 118 A: (1)(2)(3)(6)(12)

(1). Find the line passing through (1, 1) (x_1, y_1) with slope 3.

→ $y - y_1 = k(x - x_1)$

$y - 1 = 3(x - 1)$

$y - 1 = 3x - 3$

$y = 3x - 3 + 1$

$y = 3x - 2$ //

(2). Find the line passing through (-1, -2) (x_1, y_1) and (3, 2) (x_2, y_2)

→ $k = \frac{y_2 - y_1}{x_2 - x_1}$

$k = \frac{2 - (-2)}{3 - (-1)}$

$k = \frac{2 + 2}{3 + 1}$

$k = \frac{4}{4}$ | $k = 1$ //

$y - y_1 = k(x - x_1)$

$y - (-2) = 1(x - (-1))$

$y + 2 = 1(x + 1)$

$y + 2 = x + 1$

$y = x + 1 - 2$

$y = x - 1$ //

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ge 118 A (1)(2)(3)(6)(12)

(3). Find the axis of symmetry, vertex and extremum of $y = x^2 - 4x + 2$.

→ Axis of symmetry

$x = \frac{-b}{2a}$

$x = \frac{-(-4)}{2(1)}$

$x = \frac{4}{2}$ | $x = 2$ //

• vertex (顶点)

$x_p = \frac{-b}{2a} = 2$ //

Vertex = (2, −2)

$y_p = \frac{b^2 - 4ac}{-4a} = \frac{(-4)^2 - 4(1)(2)}{-4(1)}$

$= \frac{16 - 8}{-4} = \frac{8}{-4} = -2$ //

• Extremum (最值)

$y = x^2 - 4x + 2$ → the coefficient of x^2 is positive ($a = 1 > 0$) so the parabola opens upward (开口向上), the vertex is its lowest point

Min = −2 when $x = 2$

Max = —

(6). Find the maximum & minimum values of $y = -x^2 + 2x + 3$ (a=−1, b=2) on [0, 2]

→ Axis of symmetry

$x = \frac{-b}{2a}$

$x = \frac{-2}{2(-1)} = \frac{-2}{-2}$

$x = 1$ //

$x = 1$

$y = -(1)^2 + 2(1) + 3$

$y = -1 + 2 + 3$

$y = 4$ //

Vertex (顶点) = (1, 4) //

$a = -1 < 0$, so the parabola opens downward (开口向下), the vertex gives the maximum value

[0, 2] → $x = 0$, $x = 1$, $x = 2$

当 $x = 0$ 时	当 $x = 1$ 时	当 $x = 2$ 时
$y = -(0)^2 + 2(0) + 3$	$y = -(1)^2 + 2(1) + 3$	$y = -(2)^2 + 2(2) + 3$
$y = 3$ //	$y = 4$ //	$y = 3$ //

• Maximum value (最大值) = 4 when $x = 1$

• Minimum value (最小值) = 3 when $x = 0 \cup 2$ //

奇偶性

(12). Determine the parity of $x^4 + 2$, $x^3 + x$, and $x^2 + x$

→ $f(x) = x^4 + 2$

$f(-x) = (-x)^4 + 2 = x^4 + 2$

因为 $f(-x) = f(x)$ → Even function (偶函数)

$f(x) = x^3 + x$

$f(-x) = (-x)^3 + (-x)$

$f(-x) = -x^3 - x$

$f(-x) = -(x^3 + x) = -f(x)$

因为 $f(-x) = -f(x)$ → odd function (奇函数)

$f(x) = x^2 + x$

$f(-x) = (-x)^2 + (-x)$

$f(-x) = x^2 - x$

Since $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$ so the function is neither even nor odd (非奇非偶函数)

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3. $k_2 = -1$

$k_2 = -\frac{1}{3}$ //

Page 119 B (1)(2)(4)(8)

(1). Find the line passing through $(0, -1)$ and perpendicular to $y = 3x - 2$

(x_1 y_1; $k = 3$)

→ $y - y_1 = k(x - x_1)$	Perpendicular $k = -\frac{1}{3}$, passing through $(0, -1)$
$y - (-1) = 3(x - 0)$	$y - y_1 = k(x - x_1)$ → $y = -\frac{1}{3}x - 1$ //
$y + 1 = 3x$	$y - (-1) = -\frac{1}{3}(x - 0)$
$y = 3x - 1$ //	$y + 1 = -\frac{1}{3}x$

(2). A parabola has zeros $-1, 3$, and intersects the y-axis at $(0, -3)$. Find its analytic expression and vertex

(x y)

→ $x = -1$, $x = 3$	$y = (x+1)(x-3)$
$x = 1$ // $x = -3$ //	$y = x^2 - 3x + x - 3$
$y = a(x+1)(x-3)$	$y = x^2 - 2x - 3$ //
$-3 = a(0+1)(0-3)$	↳ $a = 1$, $b = -2$
$-3 = -3a$	↳ analytic expression
$a = \frac{-3}{-3}$ / $a = 1$ //	

vertex →

$x = \frac{-b}{2a} = \frac{-(-2)}{2(1)} = \frac{2}{2}$

$x = 1$ //

$y = (1)^2 - 2(1) - 3$

$y = 1 - 2 - 3 = -4$ //

Vertex $= (1, -4)$

(4) Given that the quadratic function satisfies $f(1) = f(3) = 0$, and its minimum value is -2, find $f(x)$.

→ $f(1) = 0$, $f(3) = 0$, so the zeros are $x = 1$ and $x = 3$

$x = -1$ // $x = -3$ //

$f(x) = a(x-1)(x-3)$ ←

The axis of symmetry is halfway between the two zeros:

$x = \frac{1+3}{2} = 2$ //, the vertex occurs at $x = 2$, min value is -2, so $f(2) = -2$

$x = 2$ → $-2 = a(2-1)(2-3)$

$-2 = a(1)(-1)$

$a = 2$

→ $f(x) = 2(x-1)(x-3)$

$f(x) = 2(x^2 - 4x + 3)$

$f(x) = 2x^2 - 8x + 6$ //

(8). If an odd function f satisfies $f(2) = 5$, find $f(-2)$. If an even function g satisfies $g(-3) = 4$, find $g(3)$

→ $f(x) = -f(-x)$ $\times(-1)$	$g(x) = g(-x)$
$-f(x) = f(-x)$	$g(3) = g(-3)$
$-f(2) = f(2)$	$g(3) = 4$ //
$-5 = f(-2)$ //	

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week 6 作业

P(101) 随堂练习 4.2.1 Excellent

(1) $y=3x^2-9x+1$

$a>0$ 时，用 $x=-\frac{b}{2a}$

$\Rightarrow a=3$

对称轴：$x=-\frac{b}{2a}\Rightarrow$

$\Rightarrow -\frac{-9}{2\cdot 3}=\frac{9}{6}=\frac{3}{2}$

$x=1,5$

顶点：$y=3\cdot 1,5^2-9\cdot 1,5+1\Rightarrow$

$\Rightarrow y=-5,75$

顶点坐标为：$(1,5;-5,75)$

(2) $y=-x^2+4x-1$

$a<0 \Rightarrow a=-1<0$ 开口向下

用：$x=-\frac{b}{2a}$

$x=-\frac{4}{2\cdot(-1)}=\frac{-4}{-2}=2$

最大值：~~$y_{max}=(x)^2$~~

$y_{max}=-(2)^2+4\cdot 2-1\Rightarrow$

$\Rightarrow y_{max}=-4+8-1$

$\Rightarrow y_{max}=3$

值域为 $y\le 3\ [(-\infty;3]]$

(3) $y=-x^2+4x-1$ $[0;3]$

$x=-\frac{4}{2\cdot(-1)}=2$

因为 $x=2$，在区间 $[0;3]$ 内

$\Rightarrow y_{max}=-(2)^2+4\cdot 2-1=-4+8-1=3$

$x=0$；$x=3$ $\Rightarrow$ 最小值

$x=0$；$y(0)=-0^2+4\cdot 0-1=-1$

$x=3$；$y(3)=-3^2+4\cdot 3-1=2$

最小值为 -1，最大值为 3.

$-1\le y\le 3$

(4) $y=x^2+2x-3$

$x=0 \Rightarrow y=0^2+2\cdot 0-3=-3$

$y=0 \Rightarrow x^2+2x-3=0$

$(x+3)(x-1)=0$

解得：$x_1=-3$；$x_2=1$

所以，与 x 轴的交点坐标为

$(-3;0)$ 和 $(1;0)$

(5) $x_1=1$，$x_2=3$

$y=a(x-1)(x-3)$

$(0;3) \Rightarrow (x;y)$

$\Rightarrow 3=a(0-1)(0-3)$

$\Rightarrow 3=a\cdot(-1)\cdot(-3)$

$\Rightarrow 3=3a$

$\Rightarrow a=1$

将 $a=1$ 代入并展开：

$y=1(x-1)(x-3)\Rightarrow$

$\Rightarrow y=1(x^2-3x-x+3)\Rightarrow$

$\Rightarrow y=x^2-4x+3$

(6) $y=ax^2+bx-3$ ；$x=-1$；$(-4,5)$

$x=-\frac{b}{2a}$；$x=-1$

$-\frac{b}{2a}=-1 \Rightarrow \frac{b}{2a}=1 \Rightarrow b=2a$ (方程1)

$5=a\cdot(-4)^2+b\cdot(-4)-3$

$\Rightarrow 5=16a-4b-3$

$\Rightarrow 16a-4b=8\ |:4$

$\Rightarrow 4a-b=2$ (方程2)

$b=2a \Rightarrow 4a-2a=2 \Rightarrow 2a=2\Rightarrow$

$\Rightarrow a=1$

$b=2\cdot 1=2$

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P118 A组

(3) $y=x^2-4x+2$

$x=-\frac{b}{2a}$

$x=-\frac{-4}{2\cdot1}=\frac{4}{2}=2$

$y(2)=(2)^2-4\cdot2+2 \Rightarrow$

$\Rightarrow y(2)=4-8+2=-2$

$y_{min}=-2$

(6) $y=-x^2+2x+3$

$[0,2]$

第一步

$x=-\frac{2}{2\cdot(-1)}=\frac{2}{2}=1$

$y_{max}(1)=-1^2+2\cdot1+3 \Rightarrow$

$\Rightarrow y_{max}(1)=-1+2+3=4$

第二步

当 $x=0$ 时: $-0^2+2\cdot0+3=3$

当 $x=2$ 时: $-2^2+2\cdot2+3=3$

最大值为4, 最小值为3.

(12) $x^2+2;\ x^3+x;\ x^2+x$

$f(-x)=f(x)$

$f(-x)=-f(x)$

12.1. $f(-x)=(-x)^4+2=x^4+2=f(x)$

偶函数.

12.2. $g(x)=x^3+x \Rightarrow$

$\Rightarrow g(-x)=(-x)^3+(-x)=-x^3-x=$

$=-(x^3+x)=-g(x)$

奇函数

12.3. $h(x)=x^2+x$

$h(-x)=(-x)^2+(-x)=x^2-x$

因为 $h(-x)\neq h(x)$ 且 $h(-x)\neq -h(x)$

非奇非偶函数

P119 B组

(1) $y=3x-2$; $(0,-1)$

$k_1\cdot k_2=-1 \Rightarrow k_1=3$

$k_2=-\frac{1}{k_1}=-\frac{1}{3} \Rightarrow k_2=-\frac{1}{3}$

直线解析式为: $y=-\frac{1}{3}x-1$

(2) $k_1=-1, k_2=3$; $(0,-3)$

$y=a(x-x_1)(x-x_2) \Rightarrow y=a(x+1)(x-3)$

$\Rightarrow -3=a(0+1)(0-3) \Rightarrow$

$\Rightarrow -3=-3a \Rightarrow a=1$

$y=1(x+1)(x-3)=x^2-3x+x-3=$

$=x^2-2x-3$

$\begin{cases} x=-\frac{-2}{2\cdot1}=1 \\ y=(1)^2-2\cdot1-3=-4 \end{cases}$

$\Rightarrow (1,-4)$

(4) $f(x)=a(x-1)(x-3) \Rightarrow x_0=\frac{1+3}{2}=2$

$\Rightarrow f(2)=-2 \Rightarrow (2,-2)$

$-2=a(2-1)(2-3) \Rightarrow a=2$

$f(x)=2(x-1)(x-3)=2x^2-8x+6$

8) $f(-x)=-f(x) \Rightarrow f(3)=-f(2)=-5$ 奇

$g(-x)=g(x) \Rightarrow g(3)=g(-3)=4$ 偶

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p.101 4.2.1

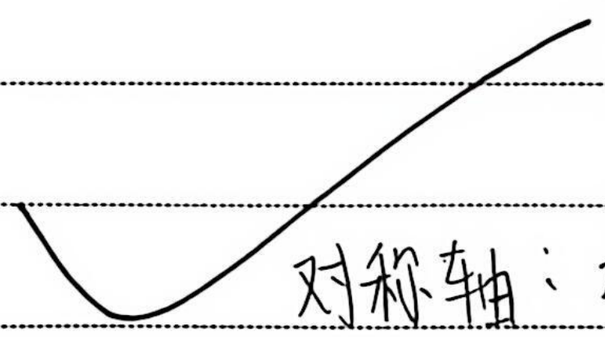
1) $a>0$，开口向上

$y=3(x^2-3x)+1$

$y=3(x-\frac{3}{2})^2-3\cdot(\frac{3}{2})^2+1$

$=3(x-\frac{3}{2})^2-3\cdot\frac{9}{4}+1$

$=3(x-\frac{3}{2})^2-\frac{23}{4}$



对称轴：$x=\frac{3}{2}$

顶点：$(\frac{3}{2},-\frac{23}{4})$

2) $y=-x^2+4x-1$

$=-(x^2-4x)-1$

$=-(x-2)^2+4-1=-(x-2)^2+3$

$(-\infty,3]$

3) $y=-x^2+4-1$

$=-(x-2)^2+3$

$y(2)=3$ $y(0)=-1$ $y(4)=2$

$[-1,3]$

4) $x=0$，$y=-3$ → $(0,-3)$

$(x+3)(x-1)=0$

$x=-3$ 或 $x=1$ → $(-3,0)$，$(1,0)$

∴ $(0,-3)$，$(-3,0)$，$(1,0)$

5) $y=a(x-1)(x-3)$

$3=a(0-1)(0-3)$

$3=3a$

$a=1$

$y=(x-1)(x-3)$

$=x^2-4x+3$

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6) $-\frac{b}{2a}=-1$

$b=2a$

$5=a(-4)^2+b(-4)-3$

$5=16a-4b-3$

$4a-b=2$

$4a-2a=2$

$a=1 \quad b=2(1)=2 \quad \therefore a=1, b=2$

P.118 3, 6, 12

3) $y=x^2-4x+2$

$=(x-2)^2-2$

最小值: $x=2$ $x=2$时取最小值 -2 开口向上 $a=1>0$,

对称轴: $x=2$ 顶点: $(2,-2)$ 最值:

6) $y=-(x^2-2x)+3$

$=-(x-1)^2+1+3$

$=-(x-1)^2+4$

$x=0, y(0)=3 \quad x=2, y(2)=3$

最大值: 4 最小值: 3

12) $f(x)=x^4+2$

$f(-x)=(-x)^4+2=x^4+2=f(x)$ 偶函数

$g(x)=x^3+x$

$g(-x)=(-x)^3+(-x)=-x^3-x=-(x^3+x)=-g(x)$ 奇函数

$h(x)=x^2+x$

$h(-x)=(-x)^2+(-x)=x^2-x$ 两个都不是

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p.119 1,2,4,8

1) $3 \cdot m = -1$

$m = -\frac{1}{3}$

$y = -\frac{1}{3}x - 1$

2) $y = a(x-(-1))(x-3)$

$= a(x+1)(x-3)$

$-3 = a(0+1)(0-3)$

$-3 = -3a$

$a = 1$

$y = (x+1)(x-3)$

$= x^2 - 2x - 3$

$y = (x-1)^2 - 1 - 3$

$= (x-1)^2 - 4$ 顶点：$(1, -4)$

4) $f(1) = f(3) = 0$

$x = \frac{1+3}{2} = 2$ $f(x) = a(x-1)(x-3)$

$a > 0$

$-2 = a(2-1)(2-3)$

$-2 = a(1)(-1)$

$a = 2$

$f(x) = 2(x-1)(x-3)$

$= 2x^2 - 8x + 6$

8) $f(-2) = -f(2) = -5$

$g(3) = g(-3) = 4$

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